PRACTICAL AI — ANALYSIS REPORT — RUN 1 — FIRST ATTEMPT

Predicting treated-water quality

Iteration 0 FIRST ATTEMPT · NO PREVIOUS RUN

GUARDRAILS · CHECKED BY THE RUN

- ✗ **Leakage: DQO-S present in the features**
- ✗ **Split: rows were randomly shuffled (time series)**
- ✓ Baseline computed first (8.39) and reported alongside
- ✓ Imputation: training-set medians only
- ⚠ Target MAE ≤ 4.5: met numerically (3.10) — **tick withheld**, guardrails failed

NEXT INVESTIGATIONS · THE SUGGESTIONS

- → Re-run with a **time-aware split**: train on the first 80% of days, test on the last 20%.
- → Remove **DQO-S** and every variable measured with the target.
- → Then compare honestly against the 8.39 baseline.

These three lines are what **run 2** executes.

Iterated from: nothing — this run opens the loop.

1 Question

Predict tomorrow's **output BOD (DBO-S, mg/L)** for the treatment plant, from today's inflow measurements and settings. Success metric: **MAE**. The baseline to beat: predict the historical mean.

2 Data

527 daily records, 12 variables, 92 missing cells (imputed with training-set medians). Schema follows the UCI Water Treatment Plant dataset (Poch 1993, CC BY 4.0); values regenerated offline for this session.

3 Features

`Q-E, PH-E, DBO-E, DQO-E, SS-E, SED-E, COND-E + DQO-S_leak`

Split: **random shuffle**. Features include **DQO-S_leak** — an output measured at the same time as the target.

4 Model & split

Linear regression. Split: **random shuffle (WRONG for time series)**.

5 Evaluation

MAE on test	**3.1 mg/L**

R^2 on test	**0.868**

✓ Human validation — to be ticked by the signatory

- ☐ The question is stated with a target and a unit
- ☐ Every feature exists at prediction time (no leakage)
- ☐ The split respects time: test days are strictly later
- ☐ The metric is reported on the test set only, next to the baseline
- ☐ Worst errors inspected; limits and uncertainty stated
- ☐ I approve this report for use within its stated limits

All runs precomputed for the session (no live computation on stage). Skill: skills/ml-analysis — swap in the real water-treatment.data with one line. Sources: UCI id 106; Sculley et al., NeurIPS 2015.

GUARDRAILS — THE NUMBERED LIST FROM PROMPT.TXT, VERIFIED BY THE RUN		
G1	PASS	Target variable and unit stated — DBO-S, mg/L
G2	PASS	Imputation applied, training-set medians
G3	**FAIL**	**FAIL** — DQO-S is present in the features; it is measured with the target
G4	**FAIL**	**FAIL** — rows were randomly shuffled; test days precede training days
G5	PASS	Baseline computed first (8.39) and printed beside every score
G6	PASS	Limits and uncertainty stated
G7	**WITHHELD**	Metric target met numerically (MAE 3.10) but **tick WITHHELD**: G3 and G4 failed
H1	HUMAN	Human sign-off — left empty by the run

Scientist’s panel — read before signing

ITERATION 0 · FIRST ATTEMPT

This run starts the loop. Its next investigations below become run 2’s instructions.

GUARDRAILS CHECKLIST

✗ **No feature measured with the outcome — DQO-S present in features**

✗ **Time-aware split — rows were randomly shuffled**

✓ Baseline computed first; scores reported next to it (8.39)

✓ Imputation: training-set medians only

✗ **Target MAE ≤ 4.5 — met numerically (3.10) but tick WITHHELD: guardrails failed**

LIMITS & UNCERTAINTY

The two failed checks invalidate the headline score. R^2 0.868 measures leakage, not skill.

NEXT INVESTIGATIONS

• Re-run with the time-aware split and DQO-S removed — this is run 2.

Sign-off (human only): approved by ____ date _________________